

Adam Lawrence

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Education

University of Illinois at Urbana–Champaign

PhD Candidate, Civil Engineering

Champaign, IL

Aug 2023 – Present

- Research focus: GPU-accelerated computational mechanics of free-surface flows and vat photopolymerization

University of Melbourne

Master of Engineering, Environmental (First Class Honours)

Melbourne, Australia

Aug 2021 – Dec 2023

- Relevant coursework: topped the class in Computational Fluid Dynamics and Remote Sensing

University of Melbourne

Bachelor of Science, Major in Environmental Engineering & Computing

Melbourne, Australia

Jun 2021

Awards

- **First Place, NIST AM Bench 2025 Photopolymerization Challenge (AM-BENCH 2025-09)**
- **AUKUS Scholar (2024)** - Highly selective PhD scholarship supporting Australian-American academic collaboration
- **Mann Scholarship (2023)** - Scholarship awarded to the highest-achieving hydraulic engineering student

Experience

Graduate Researcher (PhD)

University of Illinois at Urbana–Champaign

Champaign, IL

Aug 2023 – Present

- Developed computational models and digital-twin methods for vat photopolymerization
- Developed GPU-based dynamic adaptive meshing methods for free-surface hydrodynamic flows

Graduate Research Assistant

University of Melbourne

Melbourne, Australia

Jan 2023 – Aug 2023

- Built an OpenFOAM-based ship modeling framework to for autonomous maritime navigation

Research Intern

University of Melbourne

Melbourne, Australia

Dec 2021 – May 2022

- Analyzed ocean data to assess the impact of climate change on waters around South-East Asia

Publications and Manuscripts

- **A Multi-GPU Finite Element Framework for Vat Photopolymerization** *Manuscript in preparation*
- **Predictive FEM Modeling of Photopolymerization for the NIST AMB2025 Benchmark** *Manuscript in preparation*

Conference Presentations

APCOM-ACCM 2025

Novel Level-Set Methods for Free-Surface Flows: Introducing the Naturally Sharp-ened Formulation

Brisbane, Australia

Dec 2025

ASME IMECE 2025

GPU-Based Predictive Modeling of Photopolymerization with Applications to the NIST AM Benchmark

Memphis, TN, USA

Nov 2025

Technical Skills

Languages: C++, C, Python

Computational Modelling & Simulation: OpenFOAM, ANSYS, ParaView, HEC-RAS, PETSc

Parallel Computing: CUDA, MPI, OpenMP, HPC, Slurm, Linux, Dask